InternStudio AI Launchpad Program

Week 9 Assignment – Machine Learning Basics

Topics Covered:

- Introduction to Machine Learning
- Types of Machine Learning
- Machine Learning Workflow
- Linear Regression
- Logistic Regression (Classification)

Instructions:

- Answer all questions.
- Write explanations wherever required.
- Implement the practical questions using Python and Scikit-learn.
- Submit your Jupyter Notebook (`.ipynb`) along with screenshots of the outputs.

Section A – Machine Learning Fundamentals (4 Questions)

Q1.

What is Machine Learning? Explain how it differs from traditional programming with a suitable example.

Q2.

Differentiate between:

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning (DL)

Q3.

List and explain the four major types of Machine Learning with one real-world example for each.

Q4.

Explain the complete Machine Learning workflow from data collection to model deployment.

Section B – Linear Regression (4 Questions)

Q5.

What is Linear Regression? Explain when it should be used and give two real-world applications.

Q6.

Using the Iris dataset (or the dataset used during class), train a Linear Regression model and display:

- Model Coefficients
- Intercept

Q7.

Use the trained Linear Regression model to predict values for the test dataset. Display the first 10 actual and predicted values in a table.

Q8.

Evaluate your Linear Regression model using:

- Mean Absolute Error (MAE)

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Write a short conclusion based on the evaluation results.

Section C – Logistic Regression (Classification) (4 Questions)

Q9.

What is the difference between Regression and Classification? Give at least three examples of each.

Q10.

Predict the result (**Pass** or **Fail**) for three new students by providing your own feature values. Display the predictions and briefly explain the results.